Topic and advisor

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Senior researcher

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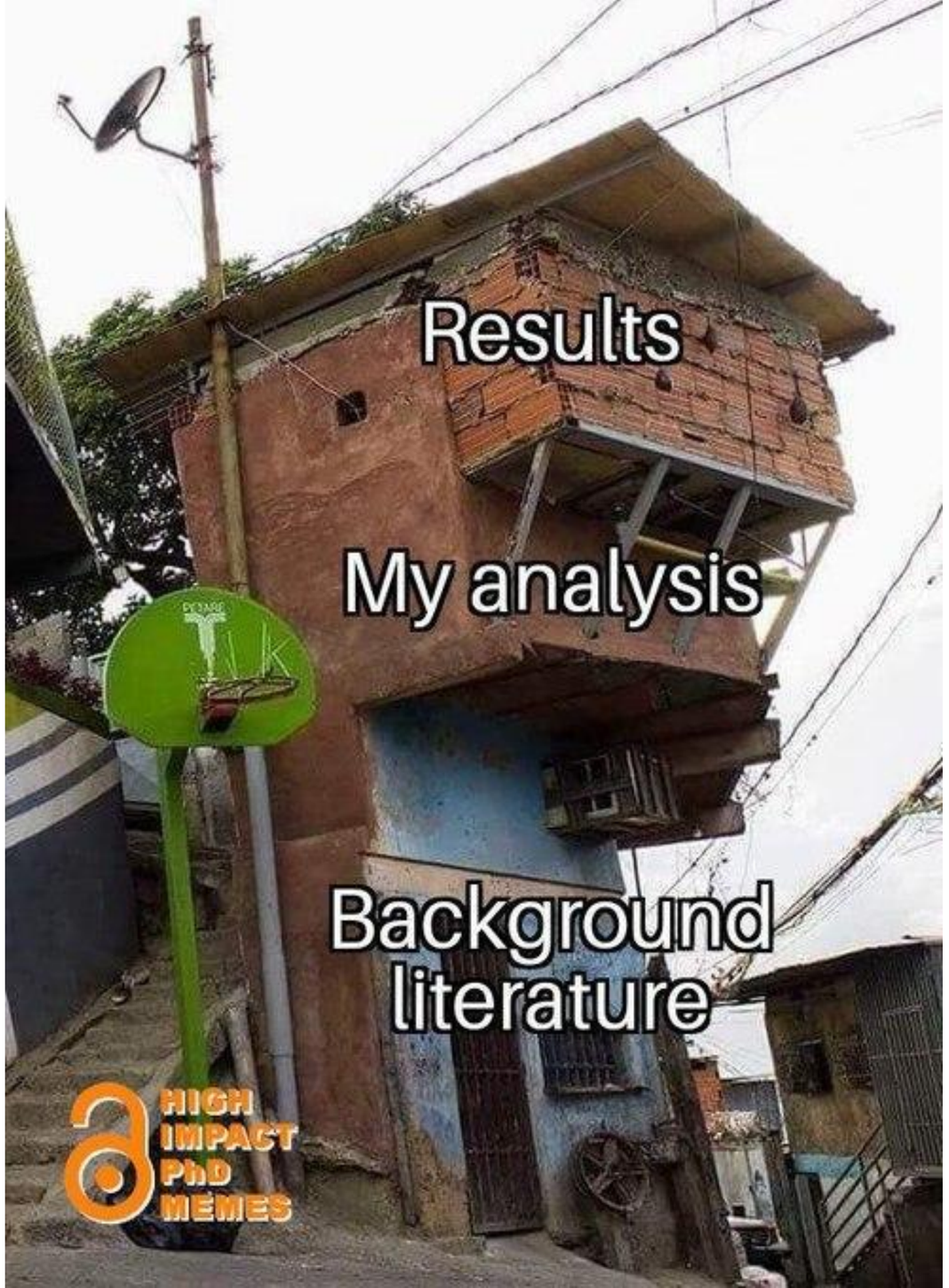
MIPT, fall 2026

Outline

1. Finding a topic
2. Finding an advisor

Steps of work

- Define a motivation
- Find a topic
- Find a thesis advisor
- Create literature overview
- Find data
- Build a baseline
- Perform experiments
- Create presentation
  - also text
- Publish a paper/poster
- Receive a diploma! 💫



Finding a topic

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01

How to choose a topic?

Sources of inspiration:

- hugging face tasks page
  - papers with code methods page (shut down)
- two minute papers
- kaggle
- google scholar
- Latest conference papers:
- **Your personal needs**
- **Your work**

- ICML
- CVPR
- NIPS
- ICCV
- ICLR
- EMNLP
- ...

Types of theses

- scientific
  - problem is fundamental
  - you develop a new method
- engineering
  - problem is concrete
  - you develop a solution for the problem

we are with you to guide from bad situations

Hype or niche?

Imagine you have to possible topics:

1. Detect cars on road
2. Detect supernova stars on telescope images

Which one to choose?

Actually answer depends on what you want.
First will be more relevant to industrial career.
Second for scientific path.

You need to consider your project value in your future life.

IMO best solution is to balance hypeness and nicheness.

Hype or niche?

1 - existing solutions
- Buisness value
- lots of tools
- lots of data

- relevant industrial exp
- wide carreer opportunities
- high competition

2. - a few solutions
- scientific value
- a few tools
- not much of data

- relevant research exp
- deep carreer ops
- softer competition

Use your existing strengths

Each choice will make you more valuable specialist, but you will narrow your field of competence.

If you have experience in some field not related to AI it’s good idea to make a project in that field:

- medicine
- physics
- chemistry
- law
- etc.

You may look at latest Nobel prizes winners. Every work includes AI.

Prepare several topics!

- you need second parachute
- finally we will help you to decide

Finding an advisor

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02

Finding an advisor

Whole thesis work is *your* qualification
and organizators' responsibility is mostly administrative.
Very promising way to find an advisor is to:

- make an overview of the topic
  - so you understand the current state of it
- compose a concise letter
  - proving you have invested your time and you have real interest
- send it to every
  - author of papers you like
  - speaker on a conference (scientific or industrial)
- ask program curators

You need to convince an advisor to work with you

Example of a thriving student

To vladislav.goncharenko@phystech.edu 24-08-2026, 07:48

Research Internship Inquiry - Real-Time Colonoscopy AI (PolypNet-3D)

Dear Prof. Goncharenko,

I have been following Your computer vision research at MIPT and how my real-time detection pipeline (PolypNet-3D) aligns with your work with great interest. Your work directly intersects with a project I have been building, and I wanted to reach out to explore potential research collaboration or internship opportunities.

I am [redacted], a pre-final year (3rd year) undergraduate in AI & Machine Learning at Easwari Engineering College, Chennai, India. I have been developing **PolypNet-3D**, a real-time clinical decision support system for colonoscopy designed explicitly for edge deployment. Key highlights include:

- A cascaded pipeline using **YOLOv8n** + **PraNet** for real-time detection and sub-pixel boundary refinement.
- A **Kalman filter** layer to resolve bounding-box flickering, achieving temporal persistence across frames.
- **TensorRT optimization** achieving 49 FPS, ensuring viability for live colonoscopy procedures on edge devices.

I am actively seeking a **research internship** (ideally with accommodation/stipend support) where I can contribute my expertise in real-time medical computer vision and edge-optimized AI architectures. I would be honored to work under your guidance at **MIPT - Computer Vision Lab, Moscow**.

I have attached my **Resume** for your reference. My portfolio and code can be viewed on my GitHub.

Thank you very much for your time and consideration. I look forward to hearing from you.

With warm regards,
[redacted]
Undergraduate Researcher, Dept. of AI & ML
Easwari Engineering College, Chennai, India

Requirements for advisors

- be able to lead you though thesis writing
  - proficiency in a field of your thesis
  - management skills and availability
  - experience in academia
    - previous successful thesis students
    - scientific articles, posters and so on

If a valid advisor (as of criteria above) is not eligible to do it formally, we will help you personally, contact support.

Thanks for attention!

Questions?



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